

Matias C.Y. Relyea

CONTACT INFORMATION	<i>E-mail:</i> hyanlong@unc.edu <i>Website:</i> matiascyrelyea.github.io
EDUCATION	University of North Carolina at Chapel Hill , Chapel Hill, North Carolina B.S. Mathematics; Minor in violin performance (expected graduation, May 2029). <i>Relevant Coursework:</i> <u>Undergraduate:</u> Differential Geometry; Topology (Duke University); Algebraic Geometry; Algebraic Structures; Advanced Calculus. <u>Graduate:</u> Modules, Linear Algebra, and Groups.
PUBLICATIONS	[1] Matias C.Y. Relyea. The Quadratic and Cubic Characters of 2. <i>Mathematics Magazine</i> . Published online January 2026. [arXiv: 2410.21646] [DOI: 10.1080/0025570X.2025.2574223].
PREPRINTS	[2] Harin Jung, Aiden Novick, Matias C.Y. Relyea. Improved Poly-Log Diameter Bounds for $SL_2(\mathbb{Z}_p)$ via Higher Commutators. <i>In preparation</i> . [3] Matias C.Y. Relyea. On Finite Fields and Higher Reciprocity (2024). [arXiv: 2407.03559].
PRESENTATIONS	[4] “Reciprocity Laws” (chalkboard lecture series). Carolina Math Club, Chapel Hill, North Carolina. March and April 2026. [5] “On Finite Fields and Higher Reciprocity” (poster). NCSSM Research Symposium Spring 2024.
RESEARCH EXPERIENCE	Einstein Institute of Mathematics , Hebrew University of Jerusalem (HUJI) REU Visiting Research Fellow, Summer 2026 Extended the Solovay-Kitaev theorem for compact Lie groups, proving new diameter bounds for $SL_2(\mathbb{Z}/p^n\mathbb{Z})$ using Elkasapy words and higher commutators. Quantum computation, Lie theory, group theory, and algebraic number theory. Mentored by Professor Shai Evra. <i>Manuscript in preparation</i>. University of North Carolina at Chapel Hill , Chapel Hill, North Carolina Directed Reading Program on Category Theory, Fall 2025 Directed Reading Program on Simplicial Homology and Cohomology, Fall 2026 North Carolina School of Science and Mathematics (NCSSM) , Durham, North Carolina Independent studies on number theory and algebraic number theory, producing one arXiv preprint and one peer-reviewed publication. August 2023—May 2025. Mentored by Professor Tamar Avineri.

Euler Circle, Mountain View, California

Attended Summer 2022 Independent Research and Paper Writing course. Independently produced 44-page expository paper on quadratic reciprocity. Mentored by Professor Simon Rubinstein-Salzedo.

SEMINARS AND [6] 25th Midrasha Mathematicae: Groups, Expanders, and Codes. Israel Institute for
CONFERENCES Advanced Studies (IAS), 14-18 June, 2026.
ATTENDED

OTHER WRITING [7] Peyton Jackson, Tatiana Medved, and Matias C.Y. Relyea. “Can You Even?” An
Exploration of Polyominoes. *NCSSM Broad Street Scientific, 2024-2025* ([link](#)).
Published online May 2026. [[PDF](#)].

[8] Matias C.Y. Relyea. Proofs and Applications of Quadratic Reciprocity. Summer
2022, *Euler Circle*. [[PDF](#)].

TEACHING AND **Mathematics Institute of the Triangle**, Chapel Hill, North Carolina
SERVICE

Mathematics Tutor and Teaching Assistant, Fall 2025—Present

- Provided one-on-one tutoring to students located in the Chapel Hill area. Assisted with grading and led short problem-solving workshops.

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Undergraduate Learning Assistant

- MATH 381 (Discrete Mathematics), Fall 2025.

Chapel Hill Math Circle, Chapel Hill, North Carolina

Program Volunteer, Fall 2025—Present

- Collaborated with UNC graduate students to assist and lead teaching for various levels of mathematics circle coursework.

TECHNOLOGICAL Proficient in $\text{\LaTeX}$, familiar with Python.
SKILLS